



Hydrological connectivity improves soil nutrients and root architecture at the soil profile scale in a wetland ecosystem

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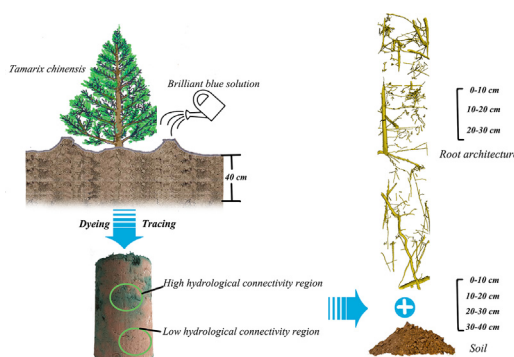
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HIGHLIGHTS

- SOC, SOM, and STN were significantly higher at 0–10 cm soil depths.
- High hydrological connectivity regions had higher SOC, SOM, and STN.
- Root length and width were obviously higher at 0–10 cm soil depths.
- High hydrological connectivity regions had higher root length, surface area, projected area, and volume.
- High hydrological connectivity promotes soil–root systems in wetland ecosystem.

GRAPHICAL ABSTRACT



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ABSTRACT

Hydrological connectivity is an essential driver of the stability, structure, and function of wetland ecosystems. Small-scale hydrological connectivity restricts large-scale hydrological cycle processes. This study aimed to investigate the response of soil and root properties to hydrological connectivity at the soil profile scale. *Tamarix chinensis*, which is a typical plant of the Yellow River Delta wetland, was sampled for analysis. We investigated soil and root properties in the three study plots where *T. chinensis* distributed and compared them at different soil depths and under different hydrological connectivity conditions. We found that the soil organic carbon (SOC), soil organic matter (SOM), and soil total nitrogen (STN) were significantly higher in shallow soil (0–10 cm deep), and that root architecture parameters such as root length and width at soil depth of 0–10 cm were also significantly higher than at other soil depths. Both the soil nutrients and root architecture parameters were significantly influenced by hydrological connectivity. Specifically, SOC, SOM, and STN were significantly higher in regions of high hydrological connectivity than in regions of low hydrological connectivity. Additionally, root length, root surface area, root projected area, and root volume were markedly higher in regions of high hydrological connectivity. Furthermore, root length and width had significant positive correlations with both SOC and SOM in regions of high hydrological connectivity. Taken together, these results indicate that higher hydrological connectivity promotes soil nutrients and root architecture at the soil profile scale. In the process of wetland protection and restoration, we should not only pay attention to hydrological connectivity at a watershed scale, but also to improving hydrological connectivity at smaller scales so as to restore soil nutrients and promote plant growth.

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1. Introduction

Hydrological connectivity refers to the process of transport of material, energy, biological matter, and other matter within or among elements of the hydrological cycle with water as the carrier (Bracken and Croke, 2007; Freeman et al., 2007; Kim and Mohanty, 2017; Masselink et al., 2016; Pringle, 2003). Different disciplines, including ecology, hydrology, geomorphology, and so on, have different definitions of hydrological connectivity. At present, there is no consensus on the definition of hydrological connectivity or how it should be measured (Bracken and Croke, 2007; Geneviève and André, 2009). Wetlands are interdependent ecosystems of vegetation, soil, and hydrology operating in the same space (Cronk and Fennessy, 2001; Zedler, 2000). As an important component of wetland ecosystems, water dominates ecological processes such as material circulation, energy flow, and biological migration; therefore, hydrological connectivity is critical for maintaining the stability, structure, and function of wetland ecosystems (Gao et al., 2017). Due to the interference of natural and human factors, wetland areas have been seriously diminished and fragmented, the hydrological connectivity within them has changed significantly, their original structure and function has been disturbed, their quality has declined, and their service function has been seriously damaged (Mitchell et al., 2013). Meanwhile, a variety of ecological and environmental problems caused by changes in hydrological connectivity have attracted attention worldwide (Dou and Cui, 2014; Liu et al., 2019a). For the above reasons, it is necessary to study hydrological connectivity in wetland ecosystems.

There are many factors affecting hydrological connectivity, including topography, soil properties, vegetation, and rainfall (Ludwig et al., 2005; Parsons et al., 2006; Wainwright and Parsons, 2002). For example, as soil organic matter content increases, the porosity of water-conducting pores first increases and then decreases; when the soil organic matter exceeds a certain threshold value, the water-conducting capacity of relatively small large pores in the soil will increase correspondingly, thus preventing the larger pores from functioning and increasing the hydrological connectivity (Larsbo et al., 2016). When the vegetation coverage is fixed, the larger in size and fewer in number the vegetation patches, the greater the hydrological connectivity (Rodríguez et al., 2017). At the slope scale, rainfall, infiltration intensity, and flow resistance are the key factors affecting runoff yield, while the connectivity of field rainfall runoff yield depends on the microtopography and the runoff resistance of the runoff path (Smith et al., 2010). The parameterization of topography and soil water content has been improved and developed to reflect hydrological connectivity (Liu et al., 2019a). Accordingly, hydrological connectivity in turn affects biodiversity (Knowlton and Jones, 1997; Lloyd and Walker, 1986), community structure (He et al., 2017; Lelieveld et al., 2015), biological diffusion (Wen et al., 2014; Xin et al., 2016), and other factors, but the influence of hydrological connectivity on these factors is controversial. For example, the biodiversity of fish has been shown to be greater in connected water than in unconnected water (Lloyd and Walker, 1986). However, the turbidity of unconnected water was shown to increase phytoplankton diversity (Knowlton and Jones, 1997). In addition, a few studies have shown that soil characteristics were affected by the strength of hydrological connectivity. There are two routes of seawater intrusion into soil: flow from the surface of the sea to inland areas via tidal creeks, and infiltration of inland soil via groundwater or soil water (Kuan et al., 2012; Mohamed et al., 2012); therefore, soil salt content is affected by the strength of hydrological connectivity (Luo et al., 2018).

Hydrological connectivity can be categorized into four dimensions: longitudinal, transverse, vertical, and temporal. The longitudinal dimension refers to the gradient from river source to river mouth; the transverse dimension refers to the connection between river and flood plain water; the vertical dimension refers to the exchange of surface water and groundwater; and the temporal dimension refers to changes in annual or long-term hydrological elements

(Ward, 1989). At present, researches on hydrological connectivity focuses on the evaluation of large-scale hydrological connectivity (Covino, 2017; Jencso et al., 2010); the relationship between hydrological connectivity and hydrological processes (Ali and Roy, 2010a; Antoine et al., 2009; Spence and Phillips, 2015); the impact of landscape pattern on hydrological connectivity (Baartman et al., 2013); the influence of anthropic factors, such as land-use types, on hydrological connectivity (Buendia et al., 2016); the assessment of hydrological connectivity in a forested hillslope by solute transport (Hanne and Harri, 2017); the hydrological connectivity structure parameters based on soil moisture content (Ali and Roy, 2010b; Antoine et al., 2009; Liu et al., 2019b) and terrain methods (Liu et al., 2019b; Liu et al., 2019c) on the plot scale; etc. It is not difficult to find that most of the researches on hydrological connectivity focus on the longitudinal and transverse dimensions on large scales such as basins, while the researches on small scales such as slope and plot scales also focus on the longitudinal and transverse dimensions. There is a lack of vertical dimension study on hydrological connectivity on small scales.

Preferential flow paths promote the rapid movement of water and solute and therefore comprehensively reflect small-scale hydrological connectivity. Preferential flow is a common form of water movement in the soil. Water and solute bypass the soil matrix area, quickly penetrate the soil layer, and even reach the underground water layer via the preferential flow path, making the water movement and the solute transport rate in the preferential flow path much higher than in the soil matrix area (Beven and Germann, 1982; Dekker and Ritsema, 1996; Hendrickx and Flury, 2001). Tracing techniques are common methods used to identify preferential flow paths in natural or constructed wetlands (Durst et al., 2013; Wanko et al., 2009). Therefore, in the present study, we used tracing techniques to determine preferential flow paths at the soil profile scale to identify areas of high hydrological connectivity.

Preferential flow can be enhanced or reduced the ability of soil to buffer and filter, which can destroy or promote ecosystem services (Clothier et al., 2008). Preferential flow is closely related to soil properties in different environments such as tropical rain forest (Noguchi et al., 1997) and paddy rice soils (Fuhrmann et al., 2019). Preferential flow path is rich in organic carbon and nutrients in forest soil (Bundt et al., 2001a) and rice soil (Fuhrmann et al., 2019), it also has higher microbial biomass in forest soil (Bundt et al., 2001b). Few studies have linked preferential flow with root characteristics. Relevant studies have simply analyzed preferential flow and root density, mainly focusing on soil characteristics (Bundt et al., 2001a). At the same time, the related researches mainly focused on forest and agricultural land, and there is little research on wetland. Therefore, this study focused on the characteristics of root-soil system including soil nutrients and root architecture in a wetland.

Despite the fact that hydrological connectivity and soil-root systems are closely interconnected, research in this area is lacking as most studies have focused on the impact of different hydrological connectivity conditions on organism and soil, not the root architecture. Existing studies on soil properties mostly concentrate on large-scale phenomena and forest ecosystem. Therefore, in the present study, we explored the influence of hydrological connectivity on soil-root systems at the soil profile scale in a wetland ecosystem.

In order to analyze the response of soil and root to hydrological connectivity at a small scale in wetlands, *Tamarix chinensis*, a plant typical of the Yellow River Delta National Nature Reserve, China, was sampled for analysis (Gao et al., 2015). Our study focused on the differences of soil nutrients and root architecture at different soil depths, the response of soil and root properties to hydrological connectivity, and the correlations between root architecture and soil nutrients. Based on research above, we provided practical guidance that may aid in understanding the impact of small-scale hydrological connectivity and in the protection and restoration of the Yellow River Delta.

2. Materials and methods

2.1. Experiment site

The Yellow River Delta National Nature Reserve is located in Dongying City, Shandong Province, China ($37^{\circ}35' - 38^{\circ}12'N$, $118^{\circ}33' - 119^{\circ}20'E$) (Han et al., 2011; Zhao et al., 2019). *Tamarix chinensis*, a typical plant of the Yellow River Delta, was selected for analysis (Gao et al., 2015). Field surveys were carried out in July 2019. Three study plots were set in flat wetland patches that were free from standing water all year round. The sample plots were $1.2\text{ m} \times 1.2\text{ m}$ to avoid border effects, and *T. chinensis* was placed at the center (Legout et al., 2009). We investigated the soil nutrients and *T. chinensis* root architecture in the rhizosphere area of the sample plots. A map of the study area and photographs of the *T. chinensis* sample plots are shown in Fig. 1.

2.2. Soil sampling and analysis

First, weeds and litter were removed from the sample plots, with care taken to minimize disturbance to the soil surface. Before dyeing and tracing, soil samples were taken from different soil depths (0–10, 10–20, 20–30, and 30–40 cm), sealed for preservation, and taken back

to measure soil nutrients, i.e., soil organic carbon (SOC), soil organic matter (SOM), soil total nitrogen (STN), and soil total phosphorus (STP), in the overall region (OR). SOC was determined by potassium dichromate method (Zhai et al., 2019; Zhao et al., 2018), SOM was determined by thermal dilution using the potassium dichromate volumetric method (Luo et al., 2018; Bai et al., 2020), STN was measured using the Kjeldtec method (Zhang et al., 2018), and STP was determined colorimetrically after wet digestion with H_2SO_4 plus $HClO_4$ (Xiao et al., 2012; Yin et al., 2020).

Brilliant blue solution was used for dyeing and tracing. After brilliant blue solution was applied, each plot was covered with plastic film to reduce rainfall erosion and surface evaporation. Twenty-four hours later, the middle part of the sample plot ($1.0\text{ m} \times 1.0\text{ m}$) was selected as the sampling area, and the soil profile was excavated and sampled (Hu et al., 2013). Dyed regions indicated preferential flow, so we regarded stained areas as high hydrological connectivity regions (HR) and unstained areas as low hydrological connectivity regions (LR) (Beven and Germann, 1982; Hanson et al., 2004; Hendrickx and Flury, 2001). Soil samples were taken from different soil depths (0–10, 10–20, 20–30, and 30–40 cm) in the dyed and unstained regions with a standard circular knife (100 cm^3). Sampling was repeated three times in the stained and unstained regions of each soil layer. The samples were stored at -2°C for analysis (Castellanos et al., 2001).

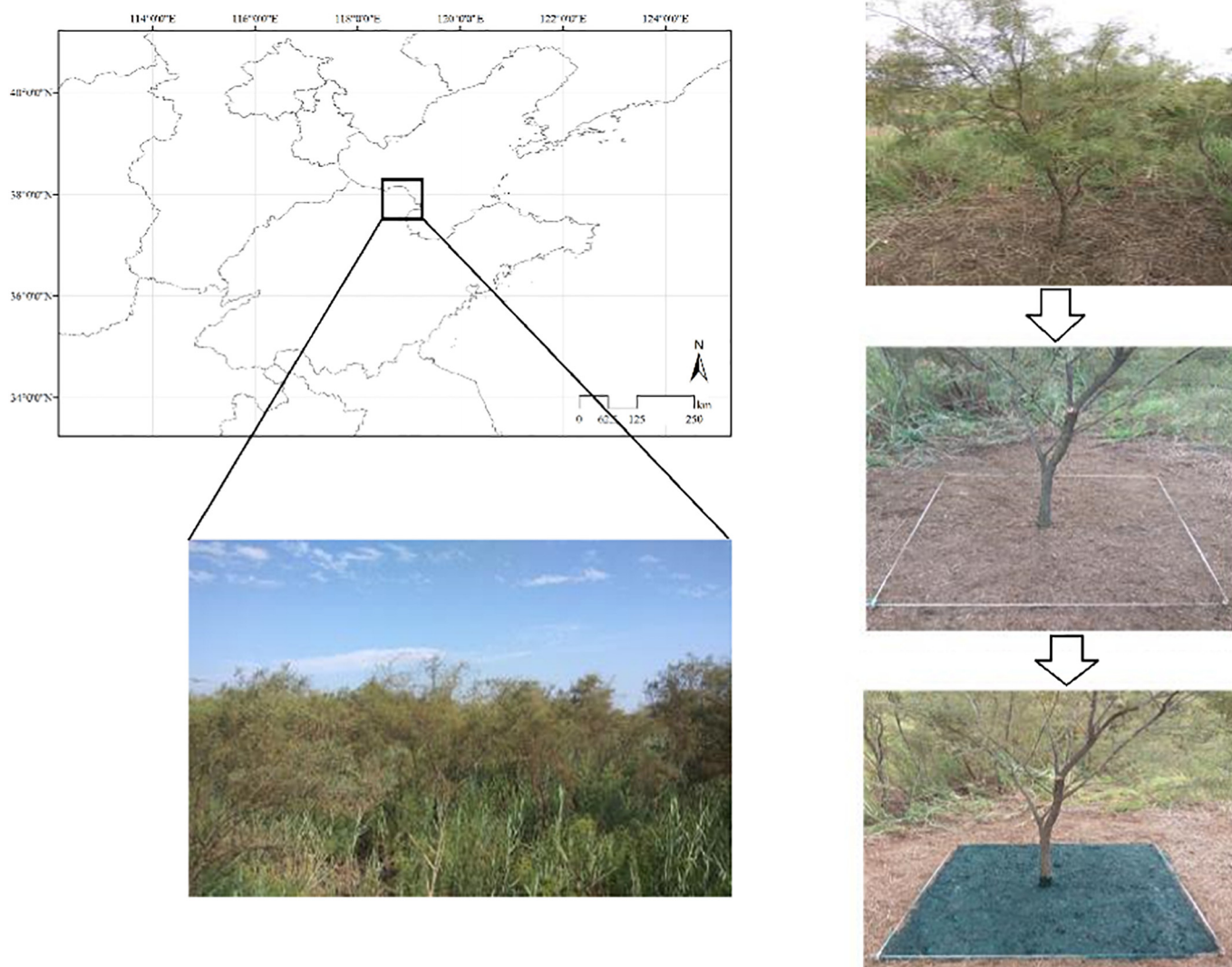


Fig. 1. Study area and sample plots.

2.3. Root sampling and analysis

The root sampling method was the same as the soil sampling method. All roots from mixed soil samples from different depths (0–10, 10–20, 20–30, and 30–40 cm) in both the HR and LR were washed to separate the roots from the soil and other particles. Care was taken to minimize damage to the root system during washing. The washed roots were dried at 70 °C for 48 h (Castellanos et al., 2001), and then root parameters, i.e., root length, root width, root surface area, root projection area, and root volume, were measured using RootAnalysis software (PM-Tech GmbH, Germany).

2.4. Statistical analysis

One-way analysis of variance (ANOVA), *t*-test, and Pearson's correlation analysis were performed using SPSS 19.0. One-way analysis of variance was performed to test the differences of soil nutrients and root architecture among different soil depths. *t*-test was used to determine the differences of soil nutrients and root architecture between HR and LR. Pearson's correlation analysis was performed to identify the relationships between soil nutrients and root architecture. The software SigmaPlot 12.5 was used for figures.

3. Results

3.1. Soil nutrients in different regions

The soil nutrients at different soil depths in three different study regions (OR, HR, and LR) are shown in Fig. 2. The SOC content around *T. chinensis* decreased as soil depth increased in all three regions. In the OR, the SOC content was 6.52 ± 1.65 g/kg at soil depth of 0–10 cm. This was higher than at other soil depths in the OR; the SOC contents were 1.61 ± 0.40 g/kg, 1.47 ± 0.62 , and 1.81 ± 0.78 g/kg at soil depths of 10–20 cm, 20–30 cm, and 30–40 cm, respectively. Meanwhile, in the HR, the SOC content decreased as soil depth increased and

was significantly higher (10.29 ± 1.89 g/kg) at soil depth of 0–10 cm than in the other three soil layers ($P < 0.01$). The SOC contents were higher at all depths in the HR than in the LR, and significantly higher at soil depth of 0–10 cm in the HR than at the same depth in the LR ($P = 0.01$). The same was true at soil depth of 20–30 cm ($P < 0.01$). In the LR, the organic carbon content was significantly higher in the shallowest soil layer (0–10 cm) than in the other three layers ($P < 0.01$).

The SOM content in the HR was higher than that in the LR. The differences in SOM content between the HR and the LR were significant at depths of 0–10 cm ($P = 0.01$) and 20–30 cm ($P < 0.01$). The SOM content at depth of 0–10 cm was higher than at all other soil depths in all three regions. In the HR, the SOM contents were 17.74 ± 3.25 g/kg, 6.60 ± 3.74 g/kg, 4.09 ± 0.48 g/kg, and 4.08 ± 0.98 g/kg at soil depths of 0–10 cm, 10–20 cm, 20–30 cm, and 30–40 cm, respectively. Compared with the above values, the SOM contents in the LR were lower and they were 12.59 ± 3.14 g/kg, 4.28 ± 1.21 g/kg, 2.94 ± 0.55 g/kg, and 2.33 ± 0.47 g/kg at soil depths of 0–10 cm, 10–20 cm, 20–30 cm, and 30–40 cm, respectively. In both the HR and the LR, the SOM contents in shallow soil (0–10 cm) were significantly higher than in other soil layers ($P < 0.01$).

In the HR, the STN content (0.93 ± 0.21 g/kg) was highest at depth of 0–10 cm ($P < 0.01$). The same was true in the LR, where the STN content was 0.64 ± 0.15 g/kg at depth of 0–10 cm, 0.22 ± 0.09 g/kg at depth of 10–20 cm, 0.12 ± 0.03 g/kg at depth of 20–30 cm, and 0.08 ± 0.03 g/kg at depth of 30–40 cm. However, the STN contents were significantly higher in the HR at depths of 0–10 cm ($P = 0.01$) and 20–30 cm ($P = 0.02$) than at the same depths in the LR. In the OR, the STN content was highest at depth of 0–10 cm (0.69 ± 0.04 g/kg) and lowest at depth of 20–30 cm (0.16 ± 0.07 g/kg).

In the OR, STP content was highest in the shallowest soil layer (0–10 cm), but only by a small margin. Unlike the soil properties above, the STP content at different depths did not differ much between the HR and the LR. However, in the LR, the STP content at depth of 10–20 cm was significantly different than that at depth of 20–30 cm

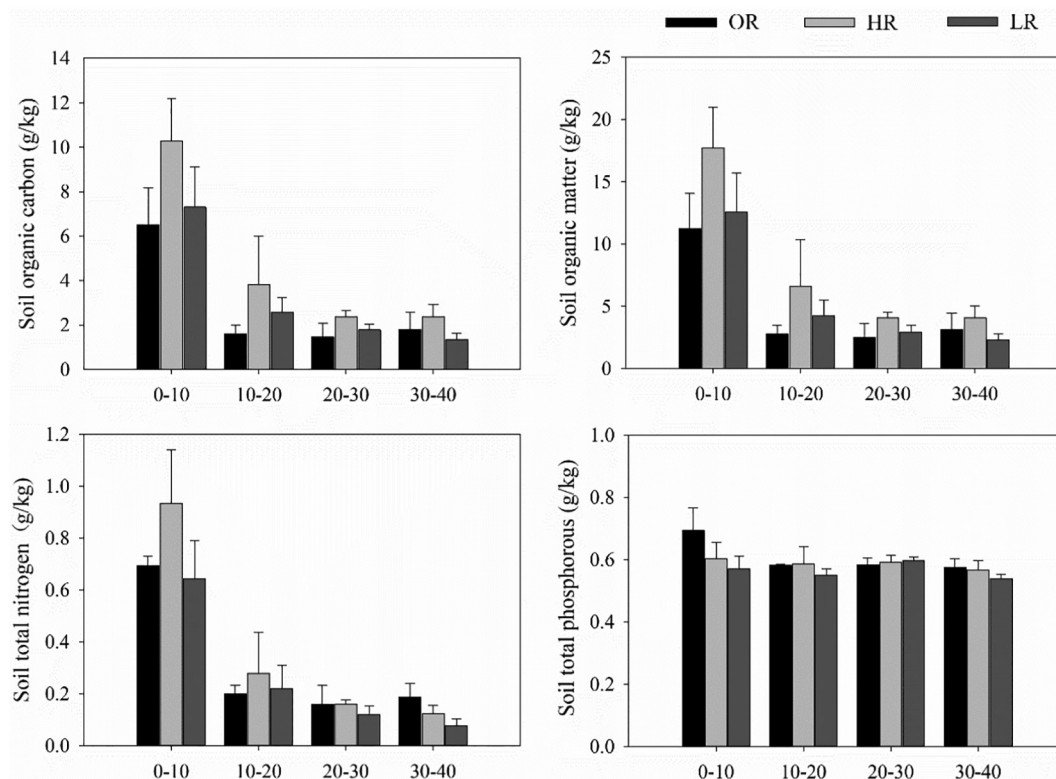


Fig. 2. Soil nutrients at different soil depths in three study regions. (OR: overall region; HR: high hydrological connectivity region; LR: low hydrological connectivity region.)

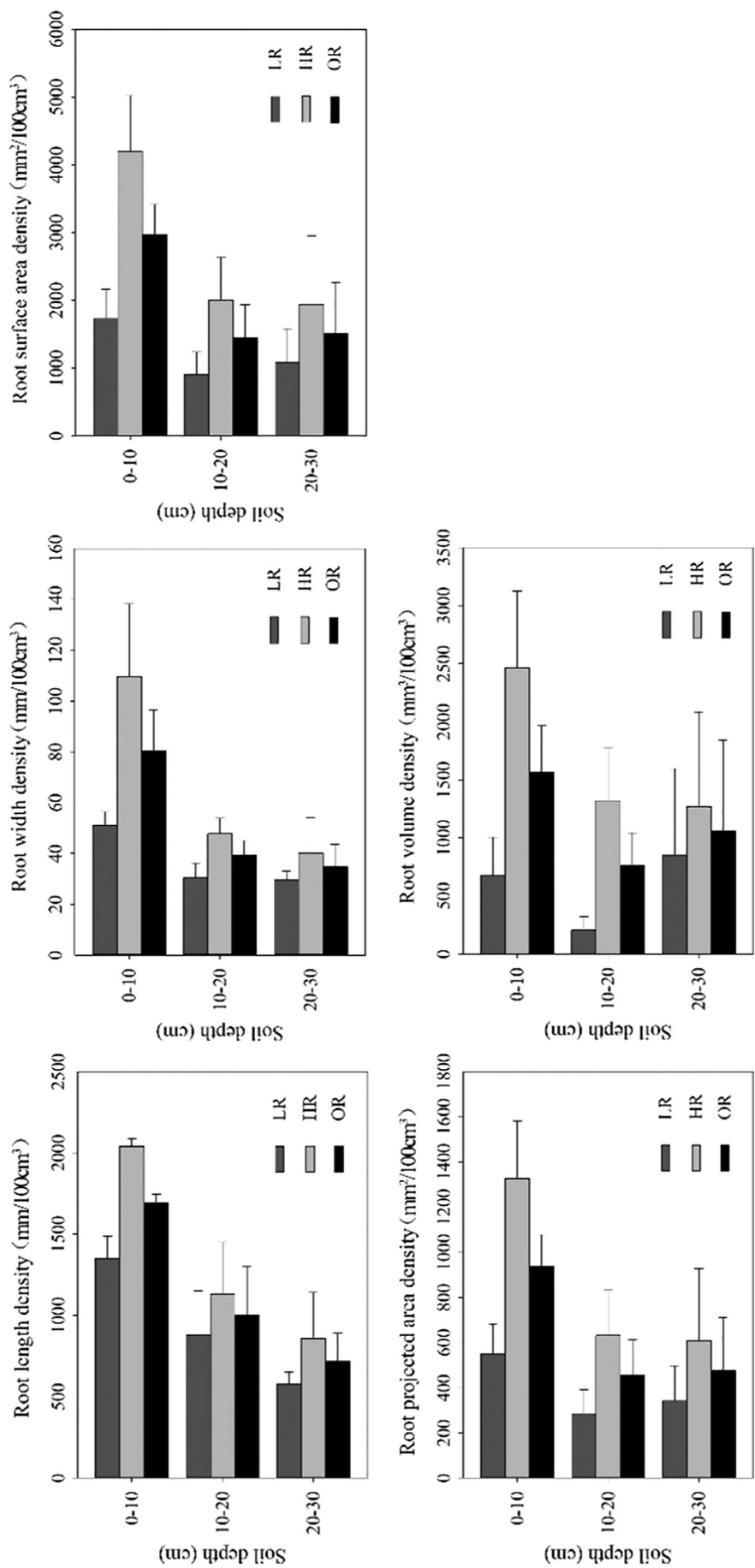


Fig. 3. Root architecture at different soil depths in three study regions. (OR: overall region; HR: high hydrological connectivity region; LR: low hydrological connectivity region.)

($P < 0.01$). The STP in the HR was 0.60 ± 0.05 g/kg, 0.59 ± 0.06 g/kg, 0.59 ± 0.02 g/kg, and 0.57 ± 0.03 g/kg at depths of 0–10 cm, 10–20 cm, 20–30 cm, and 30–40 cm, respectively; these values are higher than the corresponding values for the LR, but not significantly.

3.2. Root architecture in different regions

The values of the root architecture parameters at different soil depths in the OR, HR, and LR are given in Fig. 3. The root length density was highest in both the HR (2039.02 ± 48.83 mm/100 cm³) and the LR (1351.15 ± 133.59 mm/100 cm³) at depth of 0–10 cm. In the HR, the root length density at soil depth of 0–10 cm was significantly different from that at depths of 10–20 cm ($P = 0.04$) and 20–30 cm ($P = 0.02$). Meanwhile, in the LR, the root length density at depth of 0–10 cm was significantly different to that at depth of 20–30 cm ($P = 0.02$). The root length density in the HR different significantly from that in the LR at depth of 0–10 cm ($P < 0.01$).

The root width density also decreased with increasing soil depth in the three study regions. Root width density was higher in all soil layers in the HR than in the LR, but the differences were not significant ($P = 0.09$). In the LR, the root width density at depth of 0–10 cm differed from that at depths of 10–20 cm ($P = 0.03$) and 20–30 cm ($P = 0.03$). In the HR, the root width density at depth of 0–10 cm differed significantly from that at depth of 20–30 cm ($P = 0.04$).

In the OR, the root surface area densities were 2966.67 ± 452.38 mm²/100 cm³ at depth of 0–10 cm, 1507.56 ± 761.59 mm²/100 cm³ at depth of 20–30 cm, and 1450.25 ± 485.64 mm²/100 cm³ at depth of 10–20 cm. The root surface area densities were higher in all soil layers in the HR than in the corresponding layers in the LR ($P = 0.04$). In both the HR and the LR, the surface area density at soil depth of 0–10 cm was higher than at soil depths of 20–30 cm and 30–40 cm, but these differences were not statistically significant.

In the OR, the pattern of root projected area density was similar to that of root surface area density; at soil depth of 0–10 cm, the root projected area density was 936.23 ± 140.76 mm²/100 cm³, which was higher than in the other two soil layers. In the HR and LR, the root projected area densities in shallow soil (0–10 cm) were 1324.50 ± 256.23 mm²/100 cm³ and 547.97 ± 135.84 mm²/100 cm³, respectively; both of these values are higher than those in other soil layers. Further, the root projected area densities in the LR were significantly lower than those in the HR ($P = 0.04$).

The root volume density was highest at soil depth of 0–10 cm (1567.96 ± 395.82 mm³/100 cm³) in the OR. However, in the LR, the root volume density was highest in the 20–30 cm soil layer. In all soil layers in the HR, the root volume densities were significantly higher than in the corresponding layers in the LR ($P = 0.03$).

The remarkable differences of root architecture under different hydrological connectivity conditions indicated that hydrological connectivity of the soil significantly influenced the root architecture parameters, including the root length density, root surface area density, root projected area density, and root volume density.

3.3. Relationships between soil nutrients and root architecture

The relationships between the soil nutrients and root architecture in different soil regions are set out in Table 1. In the HR, almost all of the relationships between the soil nutrients and root architecture were positive, except for the relationship between root volume and STP. There were significant positive correlations between SOC and root length and width. SOM was also significantly positively correlated with root length and width. In addition, STN was significantly positively correlated with root width. Like in the HR, the soil properties in the LR were positively correlated with all root architecture parameters except root volume. SOC, SOM, and STN were negatively correlated with root volume in this region. However, none of the correlations in the LR were significant.

Table 1
Correlation coefficients between soil nutrients and root architecture parameters.

Region		RL	RW	RSA	RPA	RV
HR	SOC	0.677*	0.679*	0.535	0.538	0.407
	SOM	0.677*	0.679*	0.535	0.538	0.407
	STN	0.652	0.697*	0.490	0.492	0.322
	STP	0.216	0.350	0.100	0.103	−0.068
LR	SOC	0.493	0.538	0.162	0.166	−0.131
	SOM	0.470	0.514	0.122	0.126	−0.177
	STN	0.478	0.522	0.122	0.126	−0.182
	STP	−0.034	0.149	0.092	0.092	0.168

HR: high hydrological connectivity region; LR: low hydrological connectivity region.

SOC: soil organic carbon; SOM: soil organic matter; STN: soil total nitrogen; STP: soil total phosphorous; RL: root length; RW: root width; RSA: root surface area; RPA: root projected area; RV: root volume.

* $P < 0.05$.

4. Discussion

4.1. Hydrological connectivity influences soil nutrients and root architecture

Several studies have analyzed the influence of soil, vegetation, and rainfall on hydrological connectivity, and almost all at large scales (Larsbo et al., 2016; Rodríguez et al., 2017; Smith et al., 2010). However, it is also essential to analyze the effects of hydrological connectivity on these factors at small scales. In the present study, we examined areas of soil in which hydrological connectivity conditions differed in the Yellow River Delta to study the influence of hydrological connectivity on soil and root characteristics at the soil profile scale. Our results showed that both the soil nutrients and root architecture properties were significantly influenced by different hydrological connectivity conditions. Specifically, SOC, SOM, and STN in the HR were significantly higher than those in the LR. Additionally, root length density, root surface area density, root projected area density, and root volume density were markedly higher in the HR than in the LR. Thus, we can conclude that higher hydrological connectivity is beneficial for soil–root systems.

Transverse hydrological connectivity promotes the mutual flow of nutrients and energy among different wetlands and improves the capacity of wetlands to store water, retain floodwater, and purify water. Longitudinal hydrological connectivity also affects the service function of wetland ecosystems. Vertical hydrological connectivity is closely related to the mutual transformation of water, salt, and nutrients in surface water and groundwater (Clilverd et al., 2016; Obolowski et al., 2016; Zhao, 2008). In the present study, we focused on vertical hydrological connectivity at the soil profile scale; our results show differences in soil nutrients at different soil depths and under different hydrological connectivity conditions, supporting the idea that the transfer of soil nutrients is caused by hydrological connectivity. When wetland hydrological connectivity is good, the structure and composition and related physical and chemical properties of wetland soil usually remain stable. However, when hydrological connectivity is damaged or blocked, fluctuations in soil water can cause changes in wetland soil, including increases in the carbon decomposition rate and inhibition of the sedimentation process (Zhang et al., 2019).

4.2. Interactions between soil nutrients and root architecture

Roots are important organs that allow plants to absorb water and nutrients from soil. Their growth and development determine the growth dynamics of plants (Ali et al., 2018; Xu et al., 2016). Root architecture can directly reflect the growth status of plant root systems. The architecture of plant roots can not only affect their nutrient and water utilization efficiency, but can also be the basis of stable ecological communities (Xu et al., 2016). In the present study, root length, width, surface area, projected area, and volume were higher in the HR. This may be because the hydrological connectivity was higher and thus the exchange of water and nutrients was faster in this region, so the roots

Table 2
Correlation coefficients between soil nutrients.

	SOC	SOM	STN	STP
SOC	1.000			
SOM	1.000**	1.000		
STN	0.985**	0.985**	1.000	
STP	−0.174	−0.172	−0.247	1.000

SOC: soil organic carbon; SOM: soil organic matter; STN: soil total nitrogen; STP: soil total phosphorus.

** $P < 0.01$.

increased in length and surface area to absorb water and nutrients (Abenavoli et al., 2015). It has been found that when the soil nutrient composition changes, plants balance carbon distribution by adjusting the geometric parameters of their roots (Zheng et al., 2015). However, in the present study, only SOC and SOM were significantly positively correlated with root length and width in the HR. Thus, we further analyzed the relationships between soil nutrients as well as the relationships among different root architecture parameters.

The relationships between soil nutrients are set out in Table 2. There were significant positive correlations between SOC and SOM, indicating both factors may be regulated by the same processes. In addition, STN was clearly positively correlated with SOC and SOM. Unlike the above soil properties, STP was significantly negatively correlated with the other three soil properties.

The results in Table 3 show the relationships among different root architecture parameters. We found that all of the correlation coefficients among root architecture parameters were positive, and most were significant. Root length was significantly positively correlated with root width, surface area, and projected area. There were also significant positive relationships between root width, root surface area, and projected area. Unlike root surface area and projected area, neither root length nor width were significantly positively correlated with root volume.

4.3. Wetland restoration

In wetlands, hydrological connectivity is the leading factor influencing ecological processes and an important premise of wetland ecological protection and restoration (Shi et al., 2016; Zhao et al., 2016). The high intensity of human activities reduces the hydrological connection between wetlands and significantly disturbs the physical, chemical, hydrological, and biological connections between wetland patches, thus affecting the structure and function of wetland ecosystems (Kearney and Fagherazzi, 2016). Therefore, the maintenance and restoration of hydrological connectivity in wetlands is worthy of attention. As a coastal wetland, the Yellow River Delta is very sensitive and vulnerable to environmental changes and human activities.

Large-scale hydrological connectivity is an outcome of small-scale hydrological connectivity, so the restoration of small-scale hydrological connectivity is very important for wetlands. In the present study, we found that hydrological connectivity was beneficial for soil nutrients and root architecture in wetlands at the soil profile scale. Since hydrological connectivity has a significant impact on soil nutrients and plant roots at the soil profile scale, small-scale hydrological connectivity

Table 3
Correlation coefficients between root architecture parameters.

	RL	RW	RSA	RPA	RV
RL	1.000				
RW	0.810**	1.000			
RSA	0.853**	0.860**	1.000		
RPA	0.856**	0.862**	1.000**	1.000	
RV	0.603	0.658	0.919**	0.917**	1.000

RL: root length; RW: root width; RSA: root surface area; RPA: root projected area; RV: root volume.

** $P < 0.01$.

should be taken into account in wetland restoration, not only the connectivity between basins, the connectivity between upstream and downstream wetlands. Fortunately, small-scale hydrological connectivity restoration measures are easier to implement than interregional hydrological connectivity restoration measures.

5. Conclusions

Here, we quantified differences in soil nutrients and root architecture parameters at different soil depths and the response of soil and root properties to hydrological connectivity in the Yellow River Delta wetland. Our results showed that SOC, SOM, and STN were significantly higher where hydrological connectivity was high. In both the HR and the LR, all of the above soil properties were significantly higher at soil depth of 0–10 cm than at other depths. Additionally, root length, root surface area, root projected area, and root volume were markedly higher in the HR than in the LR. In the HR and the LR, root length and width were significantly higher at soil depth of 0–10 cm than in other soil layers. Furthermore, almost all the soil nutrients and root architecture parameters were positively correlated in both the HR and the LR. However, only root length and width had significant positive correlations with both SOC and SOM in the HR. In summary, we can conclude that higher hydrological connectivity is beneficial for soil–root systems in wetland ecosystems and small-scale hydrological connectivity should be considered in wetland restoration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Abenavoli, M.R., Leone, M., Sunseri, F., Bacchi, M., Sorghonà, A., 2015. Root phenotyping for drought tolerance in bean landraces from Calabria (Italy). *J. Agron. Crop Sci.* 202, 1–12.
- Ali, G.A., Roy, A.G., 2010a. A case study on the use of appropriate surrogates for antecedent moisture conditions (AMCs). *Hydrol Earth Syst Sc* 14, 1843–1861.
- Ali, G.A., Roy, A.G., 2010b. Shopping for hydrologically representative connectivity metrics in a humid temperate forested catchment. *Water Resour. Res.* 46, 65–74.
- Ali, S., Xu, Y., Ahmad, I., Jia, Q., Ma, X.C., Ullah, H., Alam, M., Adnan, M., Daur, I., Ren, X.L., Cai, T., Zhang, J.H., Jia, Z.K., 2018. Tillage and deficit irrigation strategies to improve winter wheat production through regulating root development under simulated rainfall conditions. *Agr Water Manage* 209, 44–54.
- Antoine, M., Javaux, M., Bielders, C., 2009. What indicators can capture runoff-relevant connectivity properties of the micro-topography at the plot scale? *Adv. Water Resour.* 32, 1297–1310.
- Baartman, J.E.M., Masselink, R., Keesstra, S.D., Temme, A.J.A.M., 2013. Linking landscape morphological complexity and sediment connectivity. *Earth Surf Proc Land* 38, 1457–1471.
- Bai, J.H., Yu, L., Ye, X.F., Yu, Z.B., Wang, D.W., Guan, Y.N., Cui, B.S., Liu, X.H., 2020. Dynamics of phosphorus fractions in surface soils of different flooding wetlands before and after flow-sediment regulation in the Yellow River Estuary, China. *J. Hydrol.* 580, 124256.
- Beven, K., Germann, P., 1982. Macropores and water flow in soils. *Water Resour. Res.* 18, 1311–1325.
- Bracken, L.J., Croke, J., 2007. The concept of hydrological connectivity and its contribution to understanding runoff-dominated geomorphic systems. *Hydrol Proess* 21, 1749–1763.
- Buendia, C., Batalla, R.J., Sabater, S., Palau, A., Marcé, R., 2016. Runoff trends driven by climate and afforestation in a Pyrenean Basin. *Land Degrad. Dev.* 27, 823–838.
- Bundt, M., Jäggi, M., Blaser, P., Siegwolf, R., Hagedorn, F., 2001a. Carbon and nitrogen dynamics in preferential flow paths and matrix of a forest soil. *Soil Sci. Soc. Am. J.* 65, 1529–1538.
- Bundt, M., Widmer, F., Pesaro, M., Zeyer, J., Blaser, P., 2001b. Preferential flow paths: biological ‘hot spots’ in soils. *Soil Biol. Biochem.* 33 (6), 729–738.
- Castellanos, J., Jaramillo, V.J., Sanford, R.L., Kauffman, J.B., 2001. Slash-and-burn effects on fine root biomass and productivity in a tropical dry forest ecosystem in México. *Forest Ecol Manag* 148, 0–50.

- Clilverd, H.M., Thompson, J.R., Heppell, C.M., Sayer, C.D., Axmacher, J.C., 2016. Coupled hydrological/hydraulic modelling of river restoration impacts and floodplain hydrodynamics. *River Res. Appl.* 32, 1927–1948.
- Clothier, B.E., Green, S.R., Deurer, M., 2008. Preferential flow and transport in soil: progress and prognosis. *Eur. J. Soil Sci.* 59, 2–13.
- Covino, T., 2017. Hydrologic connectivity as a framework for understanding biogeochemical flux through watersheds and along fluvial networks. *Geomorphology* 277, 133–144.
- Cronk, J., Fennessy, M.S., 2001. *Wetland Plants: Biology and Ecology*. CRC Press.
- Dekker, L.W., Ritsema, C.J., 1996. Preferential flow paths in a water repellent clay soil with grass cover. *Water Resour. Res.* 32, 1239–1249.
- Dou, P., Cui, B.S., 2014. Dynamics and integrity of wetland network in estuary. *Ecol Inform* 24, 1–10.
- Durst, R., Imfeld, G., Lange, J., 2013. Transport of pesticides and artificial tracers in vertical-flow lab-scale wetlands. *Water Resour. Res.* 49, 554–564.
- Freeman, M.C., Pringle, C.M., Jackson, R.C., 2007. Hydrologic connectivity and the contribution of stream headwaters to ecological integrity at regional scales. *J Am Water Resour As* 43, 5–14.
- Fuhrmann, I., Maarastawi, S., Neumann, J., Amelung, W., Frindt, K., Knief, C., Lehnndorf, E., Wassmann, R., Siemens, J., 2019. Preferential flow pathways in paddy rice soils as hot spots for nutrient cycling. *Geoderma* 337, 594–606.
- Gao, Z.Q., Bai, J.H., Jia, J., Wen, X.J., Zhang, G.L., Zhao, Q.Q., 2015. Spatial and temporal changes of phosphorus in coastal wetland soils as affected by a tidal creek in the Yellow River Estuary, China. *Phys. Chem. Earth* 89–90, 114–120.
- Gao, C.J., Gao, X.C., Jia, P., 2017. Summary comments on hydrologic connectivity. *J App Environ Biol (Chinese)* 23, 586–594.
- Geneviève, A.A., André, G.R., 2009. Revisiting hydrologic sampling strategies for an accurate assessment of hydrologic connectivity in humid temperate systems. *Geogr. Compass* 3, 350–374.
- Han, P., Wu, G., Wu, Y.Q., Xiang, J., Yu, L.J., 2011. Response of phragmites australis morphostructure to the different habitats in the Yellow River Delta. *Wetland Sci (Chinese)* 9, 185–190.
- Hanne, L.K., Harri, K., 2017. Model-based exploration of hydrological connectivity and solute transport in a forested hillslope. *Land Degrad. Dev.* 29, 1176–1189.
- Hanson, D.L., Steenhuis, T.S., Walter, M.F., Boll, J., 2004. Effects of soil degradation and management practices on the surface water dynamics in the Talgua River Watershed in Honduras. *Land Degrad. Dev.* 15, 367–381.
- He, F., Pan, Y., Tan, L., Zhang, Z., Li, P., Liu, J., Ji, S., Qin, Z., Shao, H., Song, X., 2017. Study of the water transportation characteristics of marsh saline soil in the Yellow River Delta. *Sci. Total Environ.* 574, 716–723.
- Hendrickx, J.M.H., Flury, M., 2001. Uniform and preferential flow mechanisms in the vadose zone. *Conceptual Models of Flow and Transport in the Fractured Vadose Zone*. National Research Council, NAP, Washington, DC, pp. 149–187.
- Hu, B., Han, C.L., Jia, Y., Zhao, Z.H., Li, F.M., Siddique, K.H.M., 2013. Visualization of the three-dimensional water-flow paths in calcareous soil using iodide water tracer. *Geoderma* 200–201, 85–89.
- Jencso, K.G., McGlynn, B.L., Gooseff, M.N., Bencala, K.E., Wondzell, A.S.M., 2010. Hillslope hydrologic connectivity controls riparian groundwater turnover: implications of catchment structure for riparian buffering and stream water sources. *Water Resour. Res.* 46, W10524.
- Kearney, W.S., Fagherazzi, S., 2016. Salt marsh vegetation promotes efficient tidal channel networks. *Nat. Commun.* 7, 12287.
- Kim, J., Mohanty, B.P., 2017. A physically based hydrological connectivity algorithm for describing spatial patterns of soil moisture in the unsaturated zone. *J Geophys Res-Atmo* 122, 2096–2114.
- Knowlton, M.F., Jones, J.R., 1997. Trophic status of Missouri River floodplain lakes in relation to basin type and connectivity. *Wetlands* 17, 468–475.
- Kuan, W.K., Jin, G.Q., Xin, P., Robinson, C., Gibbs, B., Li, L., 2012. Tidal influence on seawater intrusion in unconfined coastal aquifers. *Water Resour. Res.* 48, W02502.
- Larsbo, M., Koestel, J., Kätterer, T., Jarvis, N., 2016. Preferential transport in macropores is reduced by soil organic carbon. *Vadose Zone J.* 15.
- Legout, A., Nys, C., Picard, J.F., Turpault, M.P., Dambrine, E., 2009. Effects of storm Lothar (1999) on the chemical composition of soil solutions and on herbaceous cover, humus and soils (Fougères, France). *Forest Eco Manag* 257, 0–811.
- Lelieveld, J., Evans, J.S., Fnaiss, M., Giannadaki, D., Pozzer, A., 2015. The contribution of outdoor air pollution sources to premature mortality on a global scale. *Nature* 525, 367–371.
- Liu, J., Engel, B.A., Dai, L., Wang, Y., Wu, Y., Yan, G., Cong, L., Zhai, J., Zhang, Z., Zhang, M., 2019a. Capturing hydrological connectivity structure of wetlands with indices based on graph theory: a case study in Yellow river delta. *J. Clean. Prod.* 239, 118059.
- Liu, J., Engel, B.A., Wang, Y., Wu, Y., Zhang, Z., Zhang, M., 2019b. Runoff response to soil moisture and micro-topographic structure on the plot scale. *Sci. Rep.* 9, 2532.
- Liu, J., Engel, B.A., Wang, Y., Zhang, G., Zhang, Z., Zhang, M., 2019c. Multi-scale analysis of hydrological connectivity and plant response in the Yellow River Delta. *Sci. Total Environ.* 702, 134889.
- Lloyd, L.N., Walker, K.F., 1986. Distribution and conservation status of small freshwater fish in the river murray south Australia. *T Roy Soc South Aust* 110, 49–58.
- Ludwig, J.A., Wilcox, B.P., Breshears, D.D., Tongway, D.J., Imeson, A.C., 2005. Vegetation patches and runoff-erosion as interacting ecohydrological processes in semiarid landscapes. *Ecology* 86, 288–297.
- Luo, M., Wang, Q., Qiu, D.D., Shi, W., Ning, Z.H., Cai, Y.Z., Song, Z.F., Cui, B.S., 2018. Hydrological connectivity characteristics and ecological effects of a typical tidal channel system in the Yellow River Delta. *J Beijing Normal University (Natural Science)* 54, 17–23.
- Masselink, R.J.H., Heckmann, T., Temme, A.J.A.M., Anders, N.S., Gooren, H.P.A., Keesstra, S.D., 2016. A network theory approach for a better understanding of overland flow connectivity. *Hydrol Proess* 31, 207–220.
- Mitchell, M.G.E., Bennett, E.M., Gonzalez, A., 2013. Linking landscape connectivity and ecosystem service provision: current knowledge and research gaps. *Ecosystems* 16, 894–908.
- Mohamed, Y.A., Bastiaansen, W.G.M., Savenije, H.H.G., Van den Hurk, B.J.J.M., Finlayson, C.M., 2012. Wetland versus open water evaporation: an analysis and literature review. *Phys. Chem. Earth* 47–48, 114–121.
- Noguchi, S., Nik, A.R., Kasran, B., Tani, M., Sammor, T., Morisada, K., 1997. Soil physical properties and preferential flow pathways in tropical rain forest, Bukit Tarek, Peninsular Malaysia. *J Forest Res* 2, 115–120.
- Obolewski, K., Glińska-Lewczuk, K., Ogo, M., Astel, A., 2016. Connectivity restoration of floodplain lakes: an assessment based on macroinvertebrate communities. *Hydrobiologia* 774, 23–37.
- Parsons, A.J., Brazier, R.E., Wainwright, J., Powell, D.M., 2006. Scale relationships in hillslope runoff and erosion. *Earth Surf Proc Land* 31, 1384–1393.
- Pringle, C., 2003. What is hydrologic connectivity and why is it ecologically important? *Hydrological Proc* 17, 2685–2689.
- Rodriguez, F., Mayor, Á.G., Rietkerk, M., Bautista, S., 2017. A null model for assessing the cover-independent role of bare soil connectivity as indicator of dryland functioning and dynamics. *Ecol. Indic.* 94, 512–519.
- Shi, T., Qi, S., Meng, F., Xie, X., 2016. Land use change and landscape response in the Yellow River Delta of China: a case of Dongying City. *Environ. Earth Sci.* 75, 547.1–547.6.
- Smith, M.W., Bracken, L.J., Cox, N.J., 2010. Toward a dynamic representation of hydrological connectivity at the hillslope scale in semiarid areas. *Water Resour. Res.* 46, W12540.1–W12540.18.
- Spence, C., Phillips, R.W., 2015. Refining understanding of hydrological connectivity in a boreal catchment. *Hydrol Proess* 29, 3491–3503.
- Wainwright, J., Parsons, A.J., 2002. The effect of temporal variations in rainfall on scale dependency in runoff coefficients. *Water Resour. Res.* 38, 7–1–7–10.
- Wanko, A., Tapia, G., Mosé, R., Gregoire, C., 2009. Adsorption distribution impact on preferential transport within horizontal flow constructed wetland (HFCW). *Ecol. Model.* 220, 3342–3352.
- Ward, J.V., 1989. The four-dimensional nature of lotic ecosystems. *J. N. Am. Benthol. Soc.* 8, 2–8.
- Wen, D., Cui, B., Liu, Z., Zhang, K., 2014. Relative effects of human activities and climate change on the river runoff in an arid basin in northwest China. *Hydrol Proess* 28, 4854–4864.
- Xiao, R., Bai, J., Gao, H., Huang, L., Wei, D., 2012. Spatial distribution of phosphorus in marsh soils of a typical land/inland water ecotone along a hydrological gradient. *Catena* 98, 96–103.
- Xin, A., Sui, X., Wang, D.S., Liao, W.G., Ge, H.F., Tao, J., 2016. Landscape and avifauna changes as an indicator of Yellow River Delta Wetland restoration. *Ecol. Eng.* 86, 162–173.
- Xu, C., Tao, H., Tian, B., Gao, Y., Pu, W., 2016. Limited-irrigation improves water use efficiency and soil reservoir capacity through regulating root and canopy growth of winter wheat. *Field Crop Res.* 196, 268–275.
- Yin, S., Bai, J.H., Wang, X., Wang, X.Y., Zhang, G.L., Jia, J., Li, X.W., Liu, X.H., 2020. Hydrological connectivity and herbivores control the autochthonous producers of coastal salt marshes. *Mar. Pollut. Bull.* 160, 111638.
- Zedler, J.B., 2000. Progress in wetland restoration ecology. *Trends Ecol. Evol.* 15, 402–407.
- Zhai, J., Cong, L., Yan, G., Wu, Y., Liu, J., 2019. Influence of fungi and bag mesh size on litter decomposition and water quality. *Environ Sci Pollut R* 26, 18304–18315.
- Zhang, Y., Li, W., Sun, G., King, J.S., 2018. Coastal wetland resilience to climate variability: a hydrologic perspective. *J. Hydrol.* 568, 275–284.
- Zhang, Z.S., Yu, X.J., Song, X.L., Xue, Z.S., Wu, H.T., 2019. Impacts of hydrological connectivity on key ecological processes and functions in wetlands: a general review. *Wetland Sci* 17, 1–8 (Chinese).
- Zhao, Q.Q., Bai, J.H., Gao, Y.C., Zhao, H.X., Huang, Y.J., Zhang, W., Wang, J.N., Chen, G.H., 2019. Effects of freshwater inputs on soil quality in the Yellow River Delta, China. *Ecol. Indic.* 98, 619–626.
- Zhao, Q.Q., Bai, J.H., Huang, L.B., Gu, B.H., Lu, Q.Q., Gao, Z.Q., 2016. A review of methodologies and success indicators for coastal wetland restoration. *Ecol. Indic.* 60, 442–452.
- Zhao, Q.Q., Bai, J.H., Zhang, G.L., Jia, J., Wang, W., Wang, X., 2018. Effects of water and salinity regulation measures on soil carbon sequestration in coastal wetlands of the Yellow River Delta. *Geoderma* 319, 219–229.
- Zhao, X.B., 2008. *Study on the Hydrological Connectivity of Wetland Ecosystem*. Nanjing Hydraulic Research Institute.
- Zheng, H.L., Zhao, C.Z., Xu, T., Duan, B.B., Han, L., Feng, W., 2015. Trade-off relationship between root forks and branch angle of *Reaumuria songrica* on different aspects of slopes. *Chinese J Plant Ecol* 39, 1062–1070.